

Z11 Fabric integration validation report

HealthClaimHardened and Gateway transaction integration

PASS WITH OBSERVATIONS

Z11 adds a Fabric claim ledger flow, chaincode state controls, role checks, nullifier and challenge replay keys, and Gateway-side local verification before submit. Local chaincode and Gateway test suites pass. A deployed Fabric network was not reachable in this environment, so endorsement, ordering, commit, and end-to-end Fabric latency remain unverified.

1. Existing Fabric architecture

Repository configuration was inspected before edits. The existing topology has HospitalMSP and InsurerMSP, four peers (two per organization), three etcdraft/Raft orderers, the healthchannel channel, and the existing health chaincode. The deployed Fabric line is 2.4.9 (the compose image tags are 2.4); the CA line is 1.5.7. The existing lifecycle uses CCaaS and an AND endorsement policy requiring HospitalMSP and InsurerMSP members. Raft is crash-fault tolerant (CFT), not Byzantine fault tolerant. The topology and existing record methods were preserved.

2. Z10 to Fabric transaction flow

A caller constructs the private witness and asks HealthClaimGateway to generate a HealthClaimHardened proof. Generation locally verifies the proof and consumes its process-local challenge/nullifier. The Gateway submit API performs a second stateless proof, public-statement, request, authority, root, policy, and context check before calling Fabric. It submits only the proof envelope and claim/authority IDs. The Fabric SDK awaits a commit event and rejects non-VALID validation codes; successful results contain transaction ID, block number, claim ID, and status.

Stage	Responsibility
Proof and trust verification	Gateway / gnark Groth16
Claim lifecycle and replay state	Fabric chaincode and ledger
Endorsement / ordering / commit	Existing Fabric network; not exercised here

3. On-chain and off-chain data

The on-ledger HealthClaimProofRecord contains claim and circuit IDs, proof bytes and SHA-256 proof hash, the 13 public inputs, authority reference, submitter MSP/identity, status, explicit gateway-verification trust label, timestamp, and transaction ID. An audit record stores claim, authority/context IDs, versions, nullifier, challenge, submitter and status. The Fabric transaction does not include the private witness.

- Public inputs: registry root, root version, policy commitment, policy version, claim amount, authority public key X/Y, nullifier, protocol domain, challenge, recipient ID, deployment ID, and context commitment.
- Private/off-chain witness: diagnosis, lab value, policy ID/details, coverage ceiling, patient secret, range bounds, allowed diagnosis set, signature witness, Merkle siblings and directions, plus the raw health record.

4. Chaincode rules and trust

Registry management (trusted authority keys, accepted roots and policy commitments) requires HospitalMSP role=admin. Challenge issuance allows InsurerMSP role=insurer or zkpVerifier; proof submission requires InsurerMSP role=zkpVerifier. Claim/audit reads are role-gated; legacy GetAccessLogs is restricted to HospitalMSP admins. CreateHealthRecord is limited to HospitalMSP doctors/admins, and legacy record reads are restricted to known roles. Denied legacy consent reads no longer return the health record.

- Submission validates circuit/version, canonical BN254 field encodings and 64-bit versions/amount, claim ID, proof length (164 bytes), SHA-256 consistency, authority key reference, active root/version and policy commitment/version, challenge state/expiry/context, claim uniqueness, and nullifier uniqueness.
- The claim, nullifier key, consumed challenge, and audit are written by one chaincode invocation. On the real ledger, Fabric validation/MVCC should reject conflicting writes; the in-memory test stub does not model MVCC.
- Challenge fields bind recipient, deployment, root and policy versions; state moves ISSUED to CONSUMED. Reuse is rejected. Challenge randomness is supplied by Gateway because endorsers must execute deterministic chaincode.
- The chaincode does not run Groth16 verification, recompute Poseidon/context commitment, or independently verify proof/public-input consistency. It trusts an authorized Gateway verifier identity. A compromised or misconfigured verifier can submit a fabricated proof with matching size/hash and anchored metadata.
- Authority/root/policy registries are controlled prototype state under a HospitalMSP admin, not independently governed trust anchors. Production registry governance, key custody and rotation are unresolved.

5. Tests and performance

Reproducible local command: chaincode/scripts/z11-unit-validation.sh. Run from repository root. It runs go test -count=1 -v ./... in chaincode and go test -count=1 ./... in gateway with dedicated GOCACHE directories.

Validation	Result / scope
Chaincode contract tests	PASS: 4 top-level tests including 14 adversarial subtests; mock state only.
Gateway package suite	PASS: go test -count=1 ./... across gateway packages.
Fail-closed Gateway guard	PASS: modified invalid proof is rejected before Fabric transport call.
Live Fabric / Docker network	NOT RUN: Docker API access was denied/hung; no live peer/orderer result is claimed.
Fabric endorsement / ordering / commit latency	NOT MEASURED; no valid network sample or throughput result.
Fabric transaction loads 1/10/50/100	NOT RUN; no fabricated values.
Z9 local ZKP reference	HealthClaimHardened: 73.575 ms mean prove, about 0.7 ms verify, 164-byte proof; cryptographic-only, not Fabric latency.

Chaincode tests accept deterministic 164-byte placeholder proof bytes to exercise chaincode state logic. This is intentional evidence that chaincode acceptance depends on the authorized verifier trust boundary, not cryptographic verification. The Gateway negative test proves no transport call after local verification failure; no successful live Fabric transaction was observed.

6. Security review and unresolved issues

- Cryptographic security inherits the Z7 assumptions and parameters: Groth16 over BN254, Poseidon2, and the circuit's EdDSA/signature constraints. Fabric does not add proof soundness.
- Fabric MSP identity and endorsement configuration are relied on for caller identity and dual-organization endorsement. Raft does not protect against Byzantine orderers.

- Nullifier and challenge replay state is ledger-backed for accepted submissions, but fresh challenge issuance still depends on the Gateway challenge generator plus Fabric challenge transaction. Local challenge/nullifier memory is process-local and may be consumed even if Fabric later rejects; ledger checks remain authoritative for acceptance.
- A local challenge may remain allocated if Fabric challenge anchoring fails. Retry/recovery and cleanup are not implemented.
- The application must provision the verifier identity with the `role=zkpVerifier` attribute and the Hospital admin registry state. Existing wallets may have a verifier user without the role; enrollment migration is not automatic.
- Changes were unit-tested but not installed on a live channel. Existing sequence-1 deployments require a proper chaincode lifecycle upgrade; the initial deployment script must not be blindly rerun on an existing channel.
- Legacy record access and ZKP hash semantics remain only partially cryptographic: chaincode does not verify the legacy proof artifact; access logs expose metadata only to Hospital admins.

7. Decision

PASS WITH OBSERVATIONS

The Z11 code provides the requested state, authorization, audit, Gateway fail-closed, and replay controls in local tests. It is not validated as a live Fabric deployment. Before claiming end-to-end integration, deploy the updated chaincode under the correct lifecycle sequence, enroll the verifier attribute, populate controlled trust registries, and run real positive/negative submissions with commit-event and MVCC evidence. No Z12 work was performed.

Validation environment: Linux/amd64, Go 1.22.2 host default; gateway module selects Go 1.25.8. 16 logical CPUs; 27 GiB RAM. Measurement date: 2026-10-04. Z9 numbers are carried forward from the Z9 report and were not rebenchmarked as Fabric measurements.